# Kunal

*Machine Learning Developer*

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## Technical Skills

**Programming Languages:** Java, Python

**Frameworks:** Scikit-learn, TensorFlow, PyTorch

**Libraries:** Pandas, NumPy, Matplotlib

**Machine Learning:** Regression, Classification, Clustering, Feature Engineering, Feature Selection, Model Evaluation, LLM Evals

**Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code

**Core Concepts:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Probability & Statistics

## Projects

**Image Compression Model – K-Means Clustering Approach** | GitHub 2025

- Built an unsupervised image compression model using K-Means clustering to reduce image size by up to 90% with minimal perceptible quality loss.
- Implemented K-Means-based colour quantization to group similar pixel values, replacing them with representative cluster centroids for efficient pixel grouping.
- Designed the solution to be lightweight and computationally efficient, making it suitable for web platforms, storage optimization, and faster image transmission without heavy computational requirements.

**Jansahayak AI** | GitHub 2026

- Developed a bilingual AI assistant using Python, FastAPI, and Google Gemini API to answer government scheme-related queries in English and Hindi with source citations from 63 government documents.
- Implemented hybrid search using ChromaDB, multilingual Sentence-Transformers, and BM25 across 2,128 document chunks, improving the correct document's ranking from 8th to 1st on a benchmark query.
- Built a Hindi-English OCR document ingestion pipeline using Tesseract OCR to extract text from scanned government circulars and make previously inaccessible information searchable.
- Trained a TF-IDF and Logistic Regression classifier to categorize government documents as schemes, policies, or circulars, achieving 85% accuracy and a macro F1-score of 0.84.
- Integrated spaCy/regex-based entity extraction, LLM-generated summaries, and FastAPI REST endpoints (/ask, /documents) with a JavaScript frontend, deploying the application on Render and Netlify.

## Education

**VIT Bhopal University** Sep 2023 – May 2027

*B.Tech in Computer Science Engineering (Specialization in AI&ML)* — **CGPA: 8.6**

**Greenland Public School** Apr 2021 – Apr 2023

*CBSE* — Class XII: 77.4% — Class X: 95.4%

## Leadership & Activities

**Founder & Business Analyst, GrabKrt** – led non-technical operations (vendor onboarding, market research, outreach) for a e-commerce venture.

**PR & Outreach Member, Linux Club, VIT Bhopal** – organized sponsors and guest coordination for the university's flagship fest and maintained all the records or reports of the event.

**Volunteer, NSS Unit,** – managed crowds/donors at a blood donation camp and supported a 10,000-sapling plantation drive.

## Hackathons & Certifications

**Participant** – HackShastra (Unstop), CyberShield Hackathon (VIT Bhopal), Sandisk Hackathon (HackerEarth).

**NPTEL Cloud Computing** – Elite Certificate (IIT Kharagpur)

**Applied Machine Learning in Python** – University of Michigan